

Malt Conditioning

From German brewing and more

The crushing of the malt does not only crush the endosperm but also the husks which should remain as much as possible intact since they are needed as filter bed for lautering. However, depending on the tightness of the crush enough of the husks may or may not remain intact to allow for smooth lautering.

To alleviate lauter problems, many German brewers either condition their malt before crushing or crush their malt wet (wet crushing). The latter is also done to reduce the O₂ intake during milling and dough-in process in a modern low O₂ brewhouse, but is fairly impractical for the home brewer as it requires milling grain that is already mixed with some of the mash water.

According to a German commercial brewing magazine [Brauindustrie], basically all modern breweries that use lautertuns (as opposed to mash filters) either condition their malt or mill it wet.

Malt conditioning can be employed by the home brewer. It is however strongly recommended that this is **NOT** done with mills in home brew stores due to the risk of making a mess. In order to condition malt, the moisture content of the malt is raised by about 2% [Narziss, 2005] with the use of low temperature steam or a very fine spray of water. 2% is a very small addition of water. For a 5kg (11lb) girst this is only 100ml (3oz) of water. This moisture collects in the husks of the malt and which become much more pliable as a result of this treatment. If too much water is used for the conditioning process, it will not completely be absorbed by the husks and cause malt flour to stick to the roller of the mill. If this happens you should run some dry malt through the mill to clean it up.

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How it's done

To condition the malt put the malt in a bucket and use a spray bottle filled with water to spray the top of the malt a few times. Now mix the malt and repeat. You want to distribute the added moisture as evenly though the malt as possible. Soon the malt's feel will become less like dry straw and more like leather. Once it loses its dry feel and a few of the kernels start to stick to your hands let the malt sit for a few minutes to let the husks soak up the moisture. Get the mill ready and set it fairly tight. After all, you conditioned the malt to be able to crush it tighter. I set mine to 0.55 mm (22 mil). When you mill the malt you will notice that once in a while a crushed kernel will stick to a roller of the mill. This is ok and it only becomes a problem if a dough starts to build up on the rollers. In this case you used too much water. Run some dry malt through the mill to remove the dough.

After crushing a hand full, look at the crush. You want to see kernels that look more flattened (like oatmeal) than crushed. Pick them up and the endosperm should be dry and come out easily. If this is the case you can continue crushing the rest. If they still look more broken than flattened add more moisture and try again. In the beginning you may want to stay on the safe side and don't add too much water to avoid the risk of doughing-up the rollers.



This image shows malt samples milled with a 0.65 mm (26 mil) mill gap. The left hand sample has been conditioned and the right hand sample has been milled dry. The conditioned malt's volume is about 30% larger than the dry milled malt's volume

dry crushed malt

conditioned crushed malt



(full size)

(http://troesterfamily.com/brewing/wiki/images/9/9e/Crush_dry.jpg)



When you mill conditioned grain it will not as easily flow from the hopper into the mill. If it starts to stall, agitate the hopper or use a stick to help it along. **Do not use your fingers** especially if the mill is motor driven.

Here are pictures to illustrate the difference between the crush of dry and a conditioned malt at the same setting of the mill (JSP Maltmill, 19mil/0.48mm)

dry malt crushed



conditioned malt crushed





close-up of the same crush



close-up of the same crush

After doughing-in, you will notice that the malt does not settle as densely as dry crushed malt. This is expected as the remaining husk pieces are larger.

Conditioning Wheat Malt

Wheat malt can be conditioned as well. Though there is no husk, wheat malt still has pericarp around the endosperm, which can be preserved fairly well by malt conditioning. This process allows brewing wheat beers with a high percentage of wheat malt without the use of a lautering aid like rice hulls.

dry wheat malt crushed at 12 mil / 0.41 mm



conditioned wheat malt crushed at 12 mil / 0.41 mm





close-up of the same crush



close-up of the same crush

Common Concerns

Can conditioned malt be crushed the night before brewing?: Yes it can. The added moisture is not enough to cause spoilage.

Will it cause my mill rollers to rust?: No, it shouldn't do that. There is not enough moisture in the malt to cause rusting on the rollers. I haven't seen any and don't own a mill with stainless steel rollers.

Is it all worth it?

With the additional work and the risk of making a mess out of the mill, the question remains if conditioning the malt is worth it for the home brewer?

I think that this is an advanced technique for the home brewer and that you should not worry about it as a beginner and even a veteran should only try it when there is interest in this subject. I like to do it because it makes me feel good about my process and it does offer some advantages. Once I figured out how the malt should feel, there were no more messes with the mill either. The mash seems better suspended, which should be a good thing especially with thin mashes, but the lautering can still get stuck when using lots of grain. But that must have to do with the amount of flour that is produced with my tight crush.

Every brewer has to see for him or herself and this is definitely not a must do for excellent beer.

Sources

[Brauindustrie], Brauindustrie (http://fzarchiv.de/Zeitschriftenarchiv/Getraenke-Fachzeitschriften/Brauindustrie/1998/01_98/Bi_01-98_Aspekte_in_der_modernen_Schrotung.pdf) Technologische, qualitative und oekologische Aspekte in der modernen Schrotung, Brauindustrie Januar 1998

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